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UAE Unconventional Gas Wells Completion Operation – First Year Learning Curve During Activity Ramp-Up

A. Ryan, Z. R. Hamidon, A. Alghamdi, and C. A. Brown, Turnwell, Abu Dhabi, UAE; S. A. Elazab, S. Elhanbouly, and T. Itibrou, ADNOC, Abu Dhabi, UAE

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Abstract

Earlier in 2024, the UAE launched a transformative plan to develop its unconventional resources, focusing on cost-effective, sustainable, and technologically advanced energy production. This strategic initiative targets challenging reserves in carbonates source rocks using advanced technologies of multistage hydraulic fracturing, aiming to:

- Enhance national energy security
- Diversify energy sources
- Reduce Environmental impact
- Use untapped energy sources to support the nation's AI transformation.
- Integrate research, regulations and community engagement

This paper explores innovative, efficiency-driven horizontal wells completion methodologies and results supporting these strategic objectives. A key focus in well completion operations in unconventional resources is reducing completion cost per foot by maximizing operational efficiency and assets utilization, through working to continuously enhance the following efficiency metrics:

- Maximize pumping hours per day: through multi-well pad operations and zipper fracturing techniques
- Reduce plugs mill out times: through enhancing milling practices and coiled tubing bottomhole assembly design
- Minimize total operations time: by implementing agile project management and enhanced water and sand logistics

Post-fracturing plug milling operation applied standardized milling procedure against different types of fracturing plugs, coiled tubing string including bottomhole assembly selection, and fit for purpose fluids for high pressure gas well operations. On the operation time optimization, agile project management approach

was applied to deploy multiple operation sites for well preparation, fracturing operation, milling operation, flowback operation, and main accommodation site in Abu Dhabi desert. Sand and water logistic deliveries were revamped to transloading and bulk delivery systems. From a well construction perspective, multiple completion operation efficiency enablers were captured to be applied starting from well construction phase.

Following a year of implementing new processes, applying updated procedures, and collecting data from multiwell pads, significant efficiency improvements have been achieved. Numerous procedural refinements were adopted, and strategic directions for subsequent years have been established. The successful deployment of zipper fracturing operations, together with associated technologies such as plug-and-play gun systems, zipper manifolds, high-pressure flexible hoses, and automated split-stream fracturing operation, resulted in immediate and measurable gains—daily pumping hours increased remarkably up to 14 hours, admittedly with still room to go. Plug and perforation operations utilizing up to 14 clusters per stage were executed. Milling time also improved markedly, decreasing from 65 hours to slightly above 24 hours for lateral lengths exceeding one mile. Logistically, the UAE transitioned entirely from sandbag cutting to bulk delivery. Areas for further improvement were identified, including fracture diagnostics, eliminating operational inefficiencies, intervention challenges, and logistical difficulties; these were reviewed and will be addressed through future mitigation plans.

The learning curve led to improved gas production, with various unconventional technologies from established operations abroad boosting efficiency. Integrated project management enabled new uses of wireline plug and perforation systems, surface wellhead equipment, and automated fracturing. Additionally, logistics for water and sand delivery have undergone significant optimization as operations have scaled up.

Introduction

[Alhosani et al. \(2024\)](#) outlined a plan to advance unconventional operations in the UAE by drilling and completing 144 horizontal wells and finishing 28 drilled-uncomplete (DUC) wells over three years. The project targets Diyab gas and Shilaif oil resources, with derisking efforts split into separate batch drilling and batch completion projects phases based on location. This paper examines the completion project learning curve in the Ruwais area, which features the Diyab gas source rock formation. [Table 1](#) provides the general properties of the Diyab gas source rock.

Table 1—General subsurface information for Diyab source rock

General Well Data		
Lateral Length	~6000	ft
Formation Temperature	~330	degF
Formation Pressure	~13,000	psi
Formation Fluid	Dry Gas	
H ₂ S content	0.5-4	%
Fracture gradient	~1.07	psi/ft
Casing size	5.5 in	26 ppf

Project execution began in the Ruwais area with the goal of reaching a final investment decision concerning the gas source rocks. This was accomplished by assessing the performance of ten gas wells located on three distinct pads. The area was previously explored under a different concession in 2020, yielding satisfactory results. [Ruiz et al. \(2019\)](#) detailed operational best practices for drilling efficiency within the Diyab formation, while [Wu et al. \(2019\)](#) documented operational lessons learned during fracturing operations.

Ten unconventional gas wells were drilled across three pads, each spaced an average of 20 km apart. These wells were planned based on data from the earlier exploration well 0, which had been drilled and

completed in the same area in an earlier project phase. [Table 2](#) provides details on well configurations. The distance between surface well locations on the same pad is 10 meters.

Table 2—Project wells for unconventional gas wells.

Pad	Well	Lateral Length (ft)
NA	Well 0	3,000
Pad 1	Well A	6,000
	Well B	6,000
	Well C	4,000
	Well D	6,000
Pad 2	Well E	6,000
	Well F	6,000
Pad 3	Well G	6,000
	Well H	4,500
	Well I	5,000
	Well J	6,000

The completion project scope includes wellsite management for all rigless completion activities, such as well preparation, fracturing plug and perforation, milling, and flowback operations. Wellsite management supports rigless operations. The supporting system involves constructing site components, including water pits, waste pits, living quarters, and managing traffic. In addition to the construction works, the wellsite management work provides safety system, mechanical lifting, waste management, well suspension, and overlooking the end-to-end completion operation process.

Well preparation operations include all necessary activities prior to the fracturing plug and perforation operation. Typically, this work is intended to ensure the well is clean and has access to the formation for the upcoming procedures. Well preparation also involves setting up the wellhead surface system to accommodate a maximum pressure limitation of 15,000 psi. Depending on specific requirements, some wells may need additional interventions such as a cement integrity log and correlation run.

During the fracturing plug and perforation operation, each well was scheduled for a 30-stages hydraulic fracturing treatment to complete approximately 6,000 ft of lateral length using variable sand and water intensities. Typically, these treatments were conducted across 10 clusters and isolated by dissolvable fracturing plugs.

The objective of milling operations is to provide full wellbore access by removing fracturing plugs, thereby facilitating the cleanup of fracturing fluid during flowback. Diyab gas wells are classified as PC-4 according to coiled tubing (CT) industry standard API RP 16ST, with anticipated surface pressures exceeding 7,500 psi, necessitating robust equipment capable of withstanding such conditions. Additionally, the wells are considered sour, with H₂S partial pressures surpassing 1,000 psi due to high initial H₂S content and elevated reservoir pressure. Accordingly, CT string design, bottomhole assembly (BHA), and fluid systems must be engineered to endure these demanding conditions to ensure operational success (Alowaid et al., 2024).

Well productivity was assessed during flowback operations, which commenced approximately three months after completion of milling activities to mitigate risks associated with simultaneous operations.

Completion and Intervention Design Scope

All wells in [Table 2](#) were constructed at least a year before the gas completion project ramp up started. On the production section, these wells were constructed using 5.5" 26 ppf monobore casing on 8.5" open

hole. These wells were cemented using flexible cement and displaced by either 17 ppg water-based mud or 11.8 ppg Calcium Chloride completion brine. Each well is isolated using back-pressure valve (BPV) and 10,000 psi rated production tree. Out of these 10 wells, Well J was designed to have 15 fracturing sleeves combined with 15 stages plug and perforation on the heel section. The remaining wells were designed to have 30 stages plug and perforation completion.

On the toe sections, there are 2 types of toe completions during this project ramp up. Well A, B, C, D, E, and F were constructed using normal shoe track that requires perforation to access the formation while Well G, H, I, and J were completed using toe initiator that requires only pressure to open and access the formation. From the project perspective, considering the uncertainties of the drilling product, wellbore cleanout was conducted in all wells including the wells with toe initiator.

Wellbore cleanout was designed to clean the wellbore from potential debris or junks resulting from the drilling operations, and to drift the well with sufficient OD for the subsequent wireline tractor operation. During this project ramp up, 4.25" junk mill was used to drift the 4.548" casing ID. By having a clean and drifted wellbore, subsequent wireline tractor perforation is expected to be able to reach the target depth to do toe perforation.

The toe perforation is designed to be around 50 ft before the shoe track. During the entire project ramp up, different toe perforation configurations were tested. On the wells that have toe initiator, the well is pressurized to open the toe initiator and gain access to the formation.

Given the project-based approach of early unconventional resource development, assessing well integrity after cementing was necessary. Cement integrity logs were run under pressure: Wells A, B, C, and D were logged to total depth, while other wells were only logged on the vertical section.

The hydraulic fracturing plan was created to achieve the project's development goals. Pad 1 aimed to assess how varying sand and water levels affect fracturing and production. Wells E and F were designed to evaluate placement in zones with lower perforation density. Wells G, H, and I used standard sand and water intensities with five cluster perforations per stage, focusing on placement challenges. Well J combined cased hole plug and perforation methods with sleeve completion requiring collet activation. [Table 3](#) outlines the design for each well.

Table 3—Multistage hydraulic fracturing design

Pad	Well	Lateral Length	Number of stage	Stage isolation	Cluster per stage	Cluster length (ft)	Perforation density (spf)	Pumping rate (bpm)	Sand intensity lbs/ft	Water intensity bbls
Pad 1	Well A	6000	30	Dissolvable plug	12	1	6	100	3500	63
	Well B	6000	30	Dissolvable plug	12	1	6	100	2500	44
	Well C	4000	30	Dissolvable plug	10	1	6	100	3500	70
	Well D	6000	30	Dissolvable plug	10	1	6	100	3500	63
Pad 2	Well E	6000	30	Dissolvable plug	14	1	6	100	2500	48
	Well F	6000	30	Dissolvable plug	10	1	3	100	1800	36
Pad 3	Well G	6000	30	Dissolvable plug	5	1	3	100	2000	38
	Well H	4500	30	Dissolvable plug	5	1	3	100	2000	38
	Well I	5000	30	Dissolvable plug	5	1	3	100	2000	38
	Well J	6000	30	Dissolvable plug	5	1	3	100	2000	38

After completing the multistage plug and perforation hydraulic fracturing treatment, milling operations were designed to be completed with single run operation with the appropriate equipment. Hourglass CT string was engineered to have precise wall thickness placement to withstand high pressure while also capable of reaching the last plug on the 6,000 ft laterals. Quenched and tempered (Q&T) string was selected to maximize performance while minimizing the risk of sulfide stress cracking (SSC) as described by [Rolovic et.al \(2017\)](#) and [Galvan et. al \(2019\)](#). Two designs, first with more conservative 95,000 psi yield strength, and second with higher yield strength of 110,000-psi were utilized and evaluated to determine the ideal configuration for the wells as shown in [Table 4](#).

Table 4—CT designs used during operation.

	Design A	Design B
CT grade	QT	QT
Yield strength	95,000 psi	110,000 psi
Maximum wall thickness	0.25"	0.276"
Minimum wall thickness	0.224"	0.175"

Completing the fracturing and milling operation, well flowback operation was conducted for at least 14 days with the objective of recovering the fracturing fluids. Flowback operation is designed to maximize the amount of the fracturing fluid recovery with minimum amount of gas to be flared. At the end of each flowback operation scope, the well was connected to early production facility or suspended using various suspension procedures.

Efficiency Initiatives

[Alhosani et. al \(2024\)](#) noted that unconventional operations use innovative methods to extract hydrocarbons from source rock, differing significantly from traditional oil and gas practices. Successful project ramp-up in these operations depends on taskforces documenting lessons learned, best practices, and guidelines. To initiate this process, projects adopt insights from various basins—including Argentina, Saudi Arabia, and North America—with the aim of replicating shale operations in North America. Efficiency initiatives, drawn from regional and North American experience, are listed in [Table 5](#).

This paper grouped and discussed the implementation result of the initiatives listed on [Table 5](#). The pronounce impact of the efficiency initiatives resulted from the intervention-less toe initiation process, wireline plug and perforation efficiency, zipper fracturing operations, and CT milling optimization.

Table 5—Efficiency Initiative during project ramp up

Category	Current Practice	New Practice	Value Proposition
Well Construction – Toe initiator	Conventional shoe track with float shoe requiring toe perforation for subsequent pump down operation	Intervention-less toe access which requires toe initiator valves	Accelerating completion operations and avoid well intervention risk of CT run and tractor toe perforation
Well Construction – Cement displacement fluid	Displace cement with mud or heavy brine necessitating a dedicated CT clean-out run	Using water or lighter brine as completion fluid instead of mud.	Reducing complexity of coiled tubing wellbore cleanout and residual dirt that restricts tractor perforation
Well preparation – Cement bond log	Cement bond logs run for every well from top of cement to TD	Cement bond log on the vertical section when required only	Reducing well preparation time, reducing well intervention risks
Plug and Perforation – Pressure control equipment	Flanged connections	Quick latch wellhead connections	Improved operational efficiency and swap times, reducing HSE risk
Plug and Perforation – Running speed	Conventional grease injection wireline PCE	Greaseless wireline pressure control equipment	Faster wireline running speed that improves efficiency
Plug and Perforation – Gun system	Conventional gun system limited to 10 clusters	Modular plug and play gun-system up to 15 clusters	Reduce wireline surface time, improve fracture intensity
Plug and perforation – Plug selection	Ultimate zonal isolation	Balancing zonal isolation with milling benefit	Shorter milling time and possible simpler CT BHA.
Logistic – Water management	Trucking water from source and stored in surface tanks	Use temporary lay-flat hoses for transfer and open pits for water storage on sites	Improved water delivery volume and time, reduced traffic, and reduce emission
Logistic – Sand first mile delivery	Imported sand	Local sand from UAE sand mine	Reduce first mile sand logistic, reduce emission, improve project economics
Logistic – Sand last mile delivery	Cutting sandbags on top of sand storage	Bulk delivery using gravity or pneumatic system	Increase sand delivery capacity, reduce HSE risk
Fracturing – Surface equipment	5-1/8" 15,000 psi rated fracturing tree. Connecting 6 treating iron to each well	7-1/16" 15,000 psi rated fracturing tree with trunkline and flexible high-pressure hose	Reducing number of treating iron connections to few flanged connections.
Fracturing – Chemicals	Multiple chemicals are being used with no benchmark production capacity	Limit additives in slickwater to friction reducer and biocide and scale inhibitor as required.	Reduced logistics and improved project economics
Fracturing – Fuel distribution system	Hot refueling	Automated refueling system	Avoid HSE risk by avoiding hot refueling and improve efficiency
Plug mill out – Coiled tubing practices	Conventional approach of plug milling with multiple sweep and CT pull-tests	Enhance cleanout practices by improving BHA and fluid system, focusing on increasing annular velocity to achieve turbulent flow	Shorter milling times without compromising service quality.
Plug mill out – Extended reach tool (ERT)	Standard rotational extended reach tool used with rotor and stator configuration	Axial pulsating tool utilizing specialized flow path without moving parts	Improved reachability with less differential pressure and without elastomer failure
Plug mill out – Remote operation support	Operation to be performed solely by field crew	Introduction of remote engineer to monitor operation and generate reports	Reduce the workload of field engineers to focus on operational efficiency
Plug mill out – Surface handling equipment	Return fluid going directly to Sand Management Systems (SMS)	Introduce sacrificial line upstream of the SMS to be able to divert flow to the pit	Allow CT to continue circulation while pulling out to vertical section
Plug mill out – Surface handling equipment	Return rate calculated by physical measurement of fluid volume	Introduce inline ultrasonic flowmeter to measure return rate in real-time	Enhance decision making ability in case of any adjustment required during operation

Intervention-less well preparation

The current drilling practice to displace the cement operation is using mud or heavy brine that requires extensive wellbore cleanout operation during well preparation. Intervention-less well preparation is required to avoid the requirement of complex and extensive wellbore cleanout operation. This can be achieved by having water or light brine as cement displacement fluid and furthermore installing the toe initiator valves

on the monobore completions. With the toe initiator valve in place, the best practice is to pressurize and open the valve followed by smaller size fracturing treatment, known as toe fracturing, against toe initiator valve. In the regular well with toe perforation using wireline tractor, small size fracturing treatment against toe perforation is also recommended to remove all possible residual dirt in the casing and ease the subsequent wireline pump down plug and perforation operations.

Wireline plug and perforation efficiency upgrading

They are divided into three subcategories, i.e wireline wellhead system, wireline running speed, and bottomhole configuration. On the conventional system these three subcategories utilized flanged lubricator connection, conventional grease-injection lubricator, and conventional longer guns that are run through conventional wireline cable. These initiatives proposed to change the wireline pressure control equipment to be a quick-latch connection that minimizes time and human intervention, change grease-injection lubricator to greaseless pressure control head and utilize plug-and-play modular gun system to be run with jacketed (polymer-coated) cable.

Zipper fracturing operations for maximizing pumping hours

After concept introduction by [Wu et. al \(2021\)](#) about zipper fracturing operation against two wells by alternating the fracturing operation and wireline pump down plug and perforation, this project ramp up allowed the zipper fracturing operation evolved from the 2 wells surface zipper fracturing operation to 4 wells zipper fracturing operation using fit-for-purpose zipper fracturing equipment such as flexible high pressure hoses to replace the conventional trunkline system that is connected to 7-1/16" 15,000 psi rated fracturing tree. To operate around the clock, these high-pressure equipment systems are supported by the automated greasing system, automated refueling system, and 24-hours delivery of sand and water.

Fracturing plug milling optimization

The efficiency initiatives were also applied beyond fracturing operation. Milling optimization is crucial to maximizing the utilization of CT equipment across the project. When starting an unconventional operation, it is very common to see the coiled tubing milling operation adopt conventional approach of plug milling. Conventional milling approach in unconventional operations normally leads to multiple milling runs, frequent sweeps that consume unnecessary water and chemicals, and unnecessary additional pipe cycling that consume the pipe more than it supposed to. Part of the CT milling optimization is to deploy dissolvable plugs and accelerate the milling time.

Learning Curves

Site Preparation

Three pads within a 20 km radius were designed to share site components and infrastructure for the UAE unconventional gas project. The flat, hard terrain required minimal earth-moving during preparation. Pads were redressed, and both water and wastewater pits constructed as required. Pad 1 houses three water pits totaling 300,000 bbls to support Pads 2 and 3, which each have additional 100,000 bbls pits for buffer storage. Due to no recycling during early development, one pit was used for temporary wastewater storage before trucking to disposal.

After pad preparation, living quarters, and safety systems were installed, followed by securing water hauling and connecting transfer equipment to fracturing buffers tanks. Well preparation included attaching flowback lines and choke manifolds. Operational readiness involves pressure testing well annuli and casings. Once tests passed, a 7-1/16", 15,000 psi fracturing tree was rigged up to support toe initiator opening or wireline tractor perforation run. Alternatively, CT wellbore cleanout or any other sealed-wellbore operation such as cement bond log can be done before swapping the tree.

Delays on Pad 1 resulted from simultaneous operations and poor equipment layout, but trunkline use improved rig-up efficiency on subsequent pads. The main identified challenge was coordinating subcontractors and vendors to meet tight schedules and higher outputs than conventional projects, often with unfamiliar workflows. Learning on the gas well completion operations, all subcontractors, vendors, and service providers must agree in advance on a signed layout. Pre-rig up meeting is required for all involved parties. A typical equipment layout is shown in Figure 1.

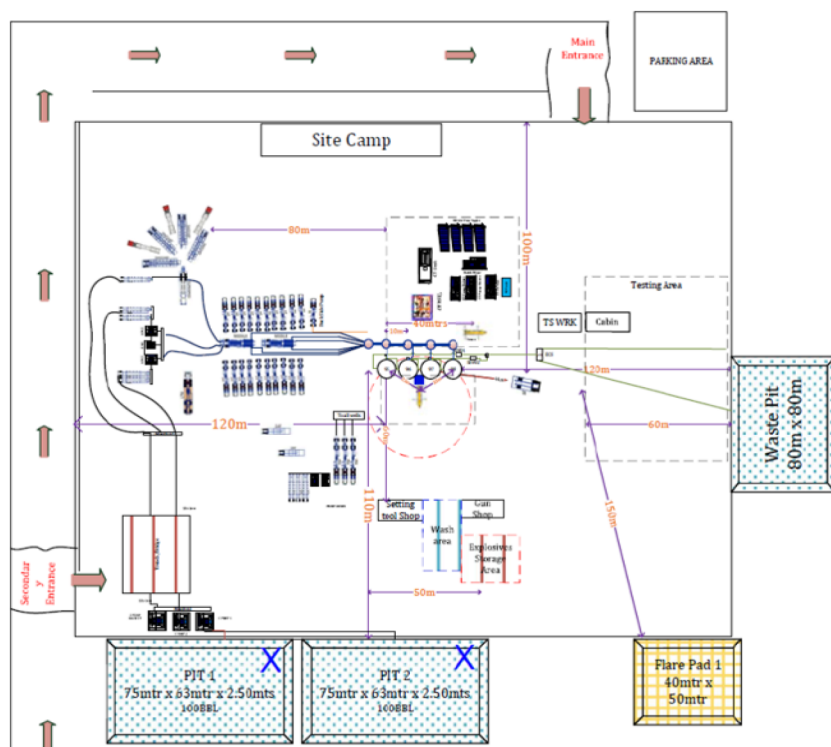


Figure 1—Typical wellsite layout for fracturing operation with its supporting system

Well Preparation

Well preparation consumed about one third of the total time of entire pad operation time. Unconventional gas completion operations captured many well preparation lessons. Some of these lessons are furthermore solved under drilling operation improvement action and some others are still pending due to resource availability. During this project, the well preparation requirement consists of CT wellbore cleanout including well drifting, cement evaluation through cement-bond log, and tractor toe perforation.

The most common and frequent inefficiency observed during gas well completion operations is multiple coiled tubing wellbore cleanout. 3 out of 10 well preparations had to redo CT cleanout operations. Two main reasons to repeat the CT cleanout operation are the excessive oil-based material on the wellbore that could not be removed by circulating out water-based fluid system and dirty wellbore condition due to solid material. Some examples of the recovered materials are shown in Figure 2.

The indication of unclean wellbore condition could only be seen when wireline is held up while running tractor to perforate. When this happens, coiled tubing needs to conduct another cleanout run before wireline can run again with tractor. This leads to additional rig up and rig down of CT and wireline units, which is time consuming and severely affects project efficiency.



Figure 2—Solid material found on coiled tubing wellbore cleanout return fluid. Left: Cement/rock debris and rubber; Middle: Metal shavings; Right: Organic material (tested in the lab)

On the completion side, some solutions identified during the project ramp up regarding the wellbore cleanliness are: 1) to install toe initiator valve on the upcoming drilling operations, 2) to conduct drift and cleanout with bigger mill size to have narrower clearance between mill and casing ID, and 3) alternative CT perforation either with CT conveyed gun system or abrasive perforation.

In some wells on Pad 1, additional depth correlation runs with Gamma-ray tool conducted against the well that is completed with pip tag instead of short joints. On the unconventional wireline plug and perforation operation, the depth correlation is conducted against the casing collar of short joints to correlate the depth efficiently. As a way forward, all the unconventional wells are using the short joints that are located before the kick-off point and at least one short joint on the lateral for wireline correlation.

On Pad 3 where the wells are completed with toe initiator valve, the opening process went as planned. This shows the toe initiator valves burst disc remained in good condition after the wells had been idle for more than a year. The subsequent learning of the toe initiator opening is to immediately displace with pump down rate as minimum rate to make sure the toe initiator port remains clean, open, and the wellbore fluid is displaced to the formation followed by toe stage fracturing treatment.

Fracturing Plug and Perforation

Multistage fracturing treatment is the main key focus during the project ramp up. The amount of equipment, people, and resources are tremendous that could lead to catastrophic failure without proper planning. Fracturing package is equipped with up to 50,000 hydraulic horsepower with the capacity up to 120 bpm at 15,000 psi maximum pressure. Supported by 10 meters distance between wells, a single wireline plug and perforation unit is utilized to support all wells in the pad. Figure 3 describes the fracturing treatment parameter ranges.

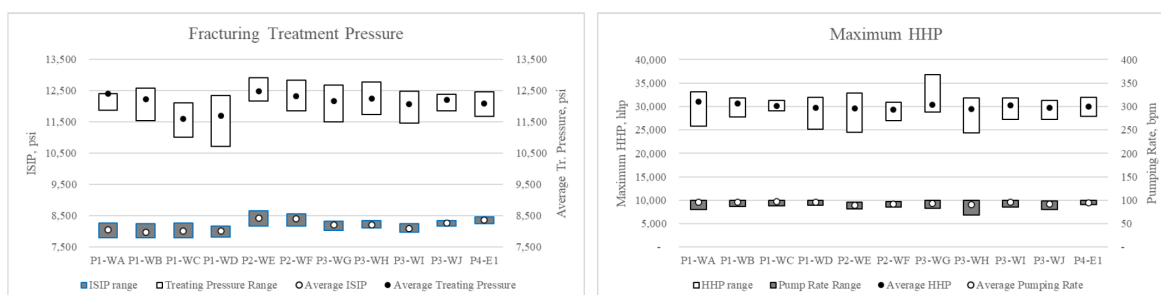


Figure 3—Fracturing Treatment Parameter. Left: Treatment Pressures. Right: Hydraulic horsepower and pumping rates

Starting from the Pad 1 execution, the average pumping hours reached 7.0 hours with maximum 14.1 pumping hours in a day. Moving forward to Pad 2 where the Well F experienced the wellhead issue that

prevented the zipper fracturing operation, the efficiency of single well fracturing treatment was at 6.9 average pumping hours per day with maximum 11 hours pumping per day. On the last pad during the project ramp up, the fracturing efficiency was at average 5.8 pumping hours per day impaired by multiple fishing operations for wireline stuck due to casing damage. The maximum pumping hours in pad 3 reached 13.6 hours per day.

Table 6 illustrates the fracturing plug and perforation operation improvement over the pads including the technology applications.

Table 6—Fracturing plug and perforation project ramp up and efficiency gain associated with the technology deployment.

	Pad 1	Pad 2	Pad 3
Fracturing Approach	zipper fracturing 2x2	single wells	4-well zipper fracturing
	split stream operation	split stream operation	split stream operation
	single wireline unit	single wireline unit	single perforation unit
Major Completion Challenge	2-well zipper fracturing	single well operation	frequent wireline stuck
	completion damage on 1 well	wellhead issues	casing deformation on several wells
Fracturing Tree Stack			multiple fishing operations
	7-1/16" 15,000 psi rated frac tree	7-1/16" 15,000 psi rated frac tree	7-1/16" 15,000 psi rated frac tree
		trunk lines	trunk lines
Plug and Perforation System			flexible high pressure hoses
	Quick latch PCE	Quick latch PCE	Quick latch PCE
	Braided Cable	Braided Cable	Jacketed Cable
	Conventional gun system	Modular gun system	Modular gun system
	12 cluster per stage, 6 spf	14 cluster per stage, 6 spf	5 cluster per stage, 3 spf
	Electrical Release Tool	Ballistic Release Tool	Ballistic Release Tool
	Dissolvable Plug	Dissolvable Plug	(Slim) Dissolvable Plug
Monitoring system	Remote job monitoring	Remote job monitoring	Remote job monitoring
Auxiliary equipment	Automated Fuel System	Automated Fuel System	Automated Fuel System
Average (max) fracturing pumping hours	7.0 (14.1)	6.9 (11.0)	5.8 (13.6)
Average rig up time	12 days	7 days	4 days
Average wireline operating time	3.5 hours	3 hours	2 hours
Average gun assembly time	8 hours	1 hour	0.5 hours
Run/Misrun ratio	10 runs	30 runs	52.5 runs
Sand last mile delivery	2.8 million lbs per day	3 million lbs per day	3.5 million lbs per day

Pad 2 deviated from the planned zipper fracturing method due to minor wellhead completion issues, leading to single-well fracturing despite supporting systems being set up for simultaneous operations. However, having multiple wells on one pad allowed continued operations without costly equipment relocation. Initial toe fracturing improved operational flow and eliminate all residual debris, that could affect subsequent pump-down operations. Wireline efficiency was enhanced with modular gun systems, quick-latch pressure controls, fit-for-purpose wireline units, jacketed (polymer-coated) cables, and crew competency upgrades, as shown in Figure 4.

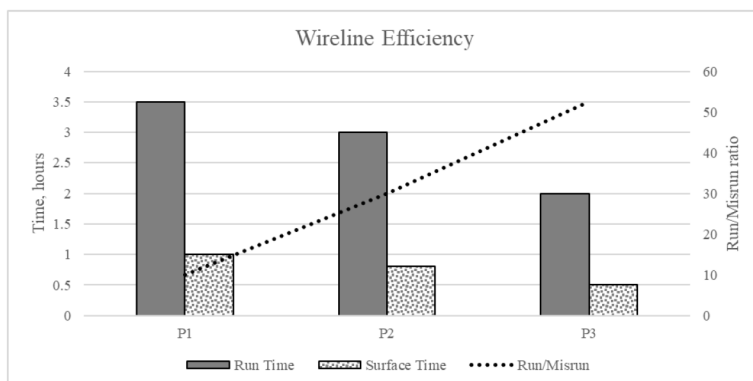


Figure 4—Wireline plug and perforation efficiency evolution.

Dissolvable plug is selected to isolate each stage for the entire 30 stages completion on each well. The plug is selected based on its ability to degrade within specific time frame at specific temperature, hence reducing the milling time. Based on the solubility test using fracturing source water, selected dissolvable plug dissolved up to 25% within 24 hours that is sufficient to lose the isolation integrity. The plug test was extended until fully dissolved after 17 days at 180 degF environment for data collection purposes. On the minimum isolation integrity verification, the dissolvable plug was tested at bottomhole condition for 8 hours against 10,000 psi which is equivalent to 3-4 times pumping time.

Slickwater was used during the project ramp up with only 2 additives, i.e friction reducer and biocide. The friction reducer additive has a tunable concentration that allows the increase the viscosity when needed. 95% of the total stages did not require viscosity adjustment for sand placement. The biocide on this application is THPS (tetrakis hydroxymethyl phosphonium sulphate).

The summary of basic completion and fracturing design are provided in Figure 5. The project started with relatively high sand and water intensity with the objective of testing the operation limit and then optimizing the design for the economic evaluations. The fracturing treatment was pumped at up to 100 bpm at more than 10,000 psi treating pressure to place sand and water with about 1.2 lb/gal placement ratio.

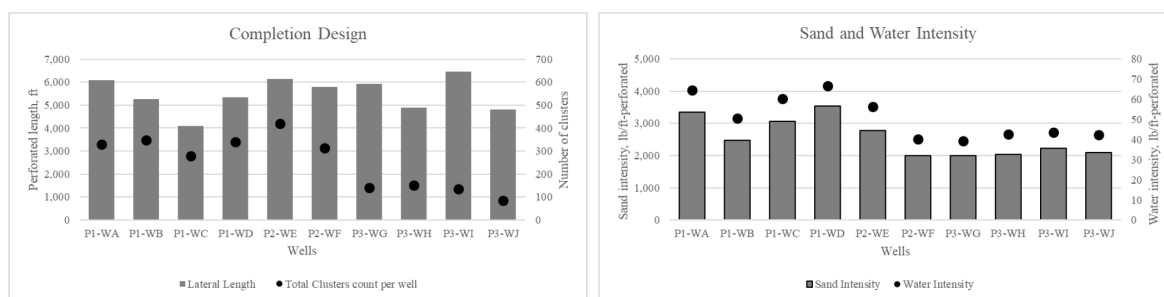


Figure 5—Left: Completion Design. Right: Sand and Water Intensity

Fracturing material and equipment logistics play significant role in maintaining operation efficiency. They also play critical role in ensuring a cost-effective operation. During the project ramp up, the two main fracturing materials, i.e. water and sand, were supposed to be delivered in bulk to wellsite. Fracturing sand was delivered to sand chiefs in 30-ton pneumatic bulkers from a transload within 20 km radius to all pads. In one pad, up to six pneumatic bulkers were offloaded parallel at the same time within 45 minutes.

During the project ramp-up, the fracturing operation efficiency did not have much improvement from pad to pad as there were different challenges that occurred in Pad 2 and Pad 3. While Pad 2 has Well F that could not be accessed for the fracturing operations and required a light workover operation, Pad 3 had multiple wireline challenges leading to many fishing intervention operations. It was recorded to have 2 pump-off incidents where the plug and perforation BHA disconnected from the wireline cable as a result of excessive forces due to possible sudden wellbore area contraction. Wireline plug and perforation BHA stuck incident was also observed on many stages in wells having same azimuth direction, i.e. 1 incident on Well G and 4 incidents on Well I.

Plug Milling Operation

CT Pipe Design

CT string for the operation was designed considering the high pressure and sour environment during milling operation:

- 2-3/8" OD was deemed to be the optimal size, to have the capability of reaching TD while maintaining operational parameter within the CT operating envelope. Bigger CT OD will have lower collapse limit, while smaller OD will limit the pump rate and the ability to reach TD.
- Q&T material with 95,000 psi yield strength was used at the beginning of the milling campaign (*CT Design A*) considering the high H₂S content of the wells. Supported by the practice of maintaining overbalance condition, comprehensive corrosion inhibition program and real-time pipe integrity monitoring, another CT design (*CT Design B*) with higher yield strength of 110,000 psi was also utilized on the latter wells in Pad 3. [Table 7](#) describes the different CT designs utilized between the 3 pads.
- Tapered CT string was precisely designed to have maximum wall thickness located at the heel section to provide stiffness, and minimum wall thickness positioned along the lateral to reduce CT weight and minimize friction. This configuration – commonly known as Hourglass design – is visualized in [Figure 6](#). Changing the material from Q&T 95,000 psi to 110,000 psi yield strengths allows for having thinner wall thickness section for CT Design B.

Table 7—CT designs used during operation.

	Design A	Design B
CT grade	Q&T	Q&T
Yield strength	95,000 psi	110,000 psi
Maximum wall thickness	0.25"	0.276"
Minimum wall thickness	0.224"	0.175"

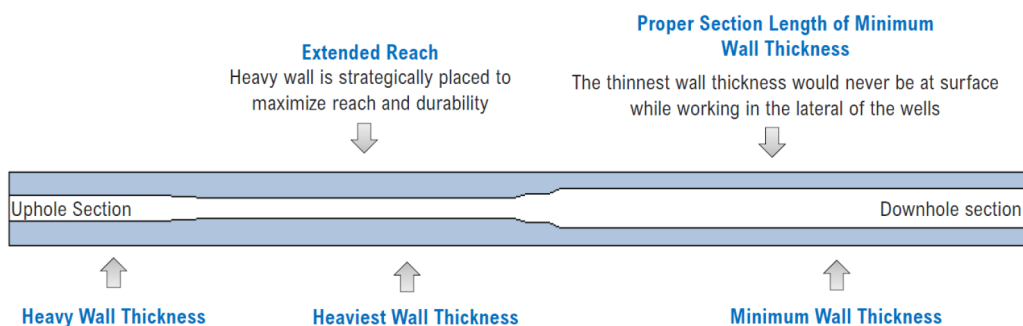


Figure 6—Wall thickness for hourglass CT string design (Alshammari et al, 2023)

Both designs of the CT string were able to reach TD in all wells without any pipe failure despite the extreme milling environment. One of the pipes reached more than 300,000 ft running length before being pulled out of service as precautionary measures.

BHA selection

BHA utilized for the milling operation consisted of CT external connector, dual check valves, hydraulic disconnect, circulation sub, extended reach tool (ERT), positive displacement motor and mill bit as shown in [Figure 7](#). The CT connector and check valves were rated to 15,000 psi as the pressure control element of the BHA. A 2-7/8" OD motor, with 4.7-stage power section capable of pumping up to 4.0 bpm was selected to be able to maximize pump rate while minimizing the circulating pressure. These criteria are especially important for the motor and ERT selection, as the high operating wellhead pressure (WHP) will lead to high CT circulating pressure which may exceed the operating envelope of the CT string.

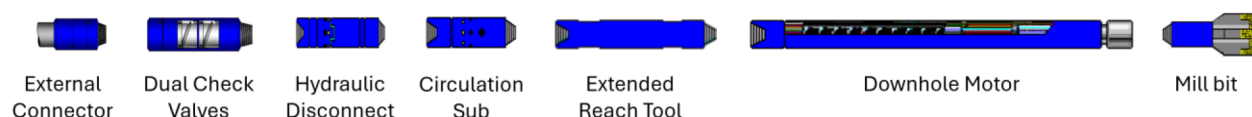


Figure 7—Standard mill out BHA Configuration

CT Connector.

- External dimple connector was used to withstand high torque application in high pressure and high temperature environment
- 15,000 psi rated to withstand high circulation pressure above 10,000 psi to achieve high pumping rate.

Dual Flapper Check Valves.

- 15,000 psi rated as part of well control requirement for PC-4 operation as per API RP 16ST Coiled Tubing Well Control Standard.

Hydraulic Disconnect.

- Ball-activated pressure-balance hydraulic disconnect.

Circulation Sub.

- Non-pressure-balanced circulation sub was used instead of burst disc, to minimize possibility of accidental burst due to the milling debris.

Extended Reach Tool (ERT).

- Axial pulsation tool was used in all wells to improve available weight-on-bit towards the toe end of the well and to reach TD.
- The selected ERT utilized specialized flow path inside the tool to generate pulsation effect. There are no elastomers in the tool, reducing potential rubber failure seen in rotational ERT.

Positive Displacement Motor (PDM).

- 2-7/8" PDM with 4.7-stage power section was used to increase the rate capability to pump up to 4.0 bpm.

Mill.

- Reverse clutch junk mill was used to have versatility on milling the different type of plugs. This mill has crushed carbide embedded into 4140 steel body.
- In wells with suspected completion damage or casing deformation, bi-center mill (also known as offset mill) was used to be able to pass through smaller ID while running-in-hole (RIH) but expand to full size during milling. This type of mill is only used as contingency, considering it might result in bigger cuttings.
- In wells with high expansion dissolvable plugs where the loose debris was hard and difficult to mill, Hurricane mill with carbide inserts were used with good results.

Other BHA component (trial): Annular Velocity (AV) Sub.

- To increase pump rate higher than the motor capability, an AV sub was trialed in one of the wells.

- AV sub with 3×0.17 " upward jetting orifice was selected based on the surface function test. The objective was to be able to achieve total 4.5 bpm pump rate, with 1.0 bpm diverted through the AV sub and 3.5 bpm going through the motor.
- During operation, the AV sub was able to achieve the objective of reaching the desired rate without significant increase in circulating pressure. However, the downhole flow diversion was found to impact motor performance, as evidenced by difficulties to mill the plugs.
- From this trial, the AV sub could be a good option for milling dissolvable plugs but may have issues milling composite plug or hard debris.

Plug selection. Dissolvable fracturing plugs were implemented with the objective of streamlining milling operations. Ensuring the compatibility of the dissolvable material with the downhole environment is essential for maintaining plug integrity during high-pressure pumping. The dissolution rates of these plugs are influenced by variables such as downhole temperature, fluid salinity, and exposure duration under downhole conditions.

Figure 8 below underscores the necessity of selecting appropriate dissolvable fracturing plugs tailored for specific downhole environments. In Pad 1 and Pad 2, a dissolvable plug previously validated for gas wells was utilized. During milling operations in both pads, the plugs had fully dissolved, thus eliminating the need for any milling intervention. However, following multiple wireline stuck incidents observed during plug and perforation procedures, an alternative dissolvable plug with a high-expansion ratio was deployed in Pad 3 to overcome restrictions. This high-expansion dissolvable plugs proved to be challenging to mill, as it did not dissolve completely. The presence of hard and loose debris led to excessive motor stalls, which caused damage to the power section. These stalls occurred not only at the plug depths but also between stages, as debris accumulated along the lateral rather than being effectively milled and circulated out of the wellbore.

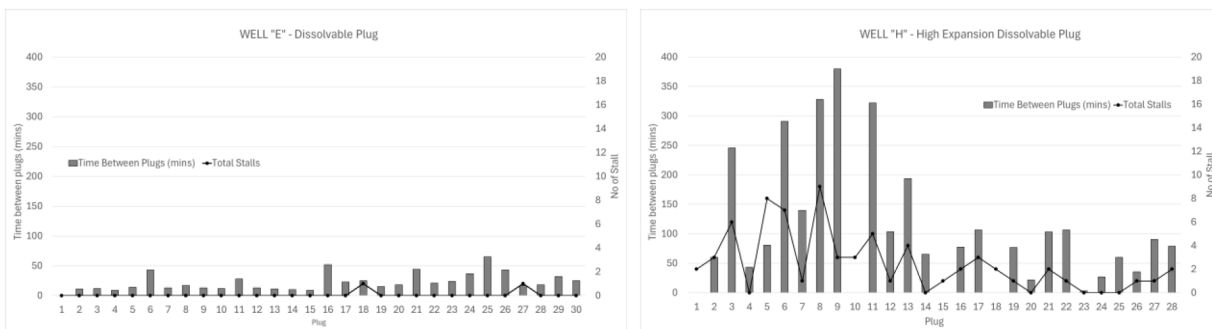


Figure 8—Comparison between the suitable dissolvable fracturing plug and contingency high expansion dissolvable plug

Figure 8 highlights the significant difference in milling performance between Well E of Pad 2 and Well H of Pad 3. Well H, equipped with high-expansion plugs, experienced considerably longer intervals between plugs and frequent motor stalls. Furthermore, the increased number of stalls resulted in premature BHA failures, necessitating multiple CT runs to complete the well.

Surface well test collaboration. It is important to establish clear communication between CT and surface well test crew and not isolating the two activities from one another. Milling operation requires collaborative effort as any miscommunication may compromise the operation. Important tasks such as choke management must be clearly defined and agreed to avoid losing circulation which may lead to CT stuck situation. Comprehensive flow chart was developed with required actions from both CT and surface well test side in case of contingency to ensure synchronized actions can be performed.

Milling in sour gas environment with potential of having H_2S at surface, all return handling equipment must be H_2S rated. All equipment upstream of the choke must be rated to 15,000 psi to be able to handle the

high pressure. Milling operation was performed with slight overbalance condition, to avoid any gas and H_2S influx from the formation. The overbalance condition is controlled by maintaining return rate slightly lower than CT pump rate, with return rate physically monitored by using surge tank. During Pad 3 operation, an inline clamp-on flowmeter was introduced to have a real-time reference for the return rate.

Solid was recovered in two stages, with plug and trash catcher (PTC) to handle plug debris, and cyclonic desander to collect sand. In case of any leaks or failure upstream of the solid management system, sacrificial lines were added to allow CT to continue circulation while pulling out of hole (POOH) to vertical section to fix the issue. Example of a return handling equipment is illustrated in Figure 9.

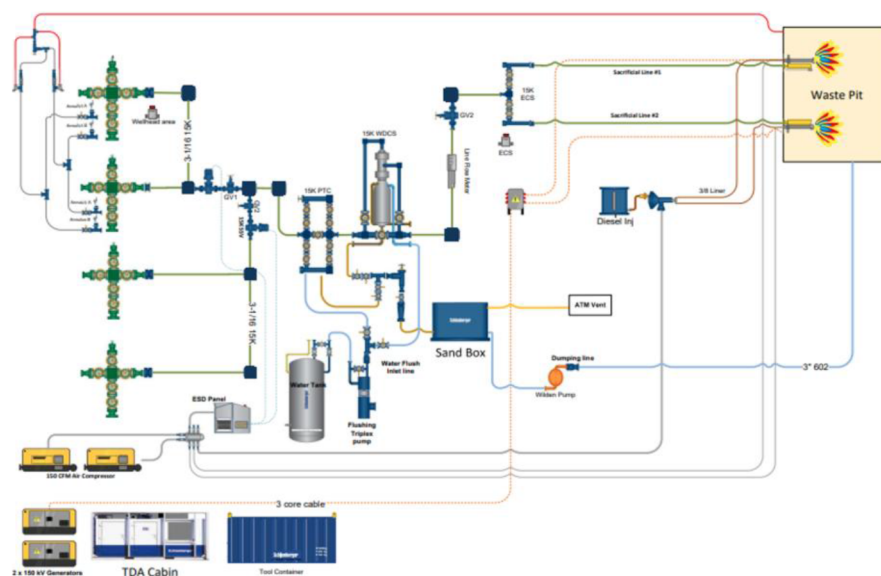


Figure 9—Milling support equipment package

Plug milling efficiency. At the start of the ramp up, a clear target was defined as the baseline for the plug milling operation. Historically, milling operation in wells with 15 plugs could take more than 3 days, in some cases with multiple runs. This project is targeting to complete milling operation, generally having 30 plugs per well, within 30 hours. This duration is measured from opening the well for the first time until closing the well after milling all plugs and reaching TD. This target is achievable if the operation is completed within a single run, while optimizing operating parameters to eliminate short or long sweeps. Reducing the time spent downhole minimizing the exposure of CT to the high-pressure sour environment, protecting CT pipe and surface equipment, reduced fluid consumptions, and minimizing overall operation risks.

Figure 10 demonstrates a comparison in milling performance between Well A, which was the first mill out for this ramp up, and Well E which is one of the fastest milling operations in this campaign. Both wells have 30 dissolvable plugs, without any known completion issue that may hinder milling efficiency. In Well A, a short sweep of 1,000 ft was performed after every 5 plugs, while there were 2 long sweeps to vertical performed due to equipment issue (after plug 2 and after plug 12 as shown in Figure 10). These events contributed to overall operating time of 64.6 hours. As the sweeps did not show any significant improvement in solid recovery, the strategy was then shifted to increase circulation rate to achieve turbulence flow and eliminate the unnecessary sweeps. As a result, the operation was able to be completed within less than 30 hours, as shown for Well E below.

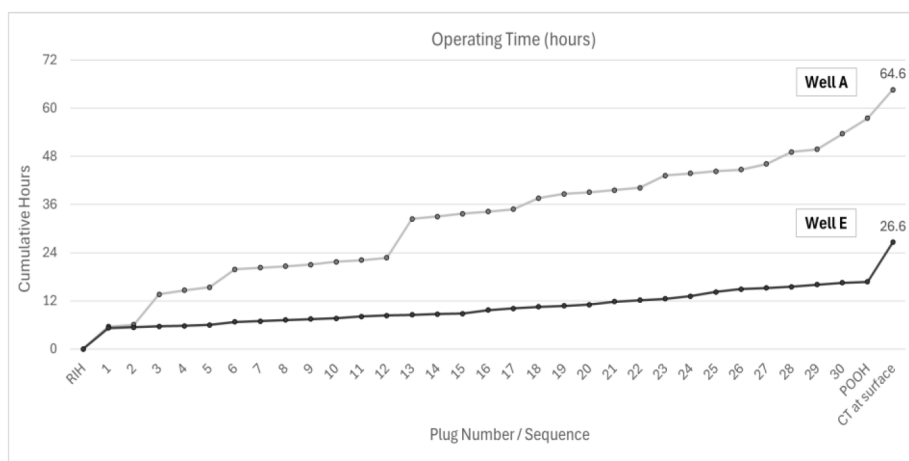


Figure 10—Plug mil lout operating time improvement

Flowback Operation

The production data presented in this paper is intended to illustrate the project scale during the final phase of the workflow. At the time of this paper's completion, only a portion of the wells in Pad 1 had been tested and subsequently connected to the early production facility. All flowback operations extended beyond the expected 14-days cleanup period to monitor individual performance. Gas hydrocarbons were observed during all flowback operations. The following operational details were noted during flowback:

- The equipment setup was adjusted to be more efficient and reduce the site footprint.
- Solids handling equipment setups and layouts were designed for flexibility and adaptability, particularly for continuous operations involving well preparation, fracturing plug and perforation, milling, and flowback processes. One PTC was used during well preparation, while none were used during fracturing operations. Desanders and sand filters were applied during milling and flowback.
- An H₂S stripper column was implemented to remove H₂S from flowback fluids prior to disposal.
- High H₂S concentrations were observed at the start of flowback operations, coinciding with lower total gas production rates, and decreased as hydrocarbon gas rates increased.
- A heat-exchanger maintained surface fluid temperature within equipment limitations.
- Multiple flow diversions occurred at the early production facility during commissioning. Flexibility in surface well test equipment allowed for diversion and flaring using on-site equipment when the production facility required adjustments.
- Strategic chemical injection points were established to address variable outcomes from the H₂S stripper column, ensuring that no H₂S remained in the flowback water.

Conclusion and Way Forward

In summary, the first year of unconventional gas well completions in the UAE saw positive progress and increased efficiency as operations ramped up. Key achievements include:

- Enhancement of the multistage plug-and-perforation hydraulic fracturing system with full supporting infrastructure.
- Notable improvements in operational efficiency metrics such as fracturing pumping hours, wireline runs, mobilization, and coiled-tubing milling times.
- The early production performance of the wells has contributed to the final investment decision (FID) sanction process and field development plan (FDP) processes.

- Deployment of advanced technologies improved both efficiency and completion quality, successfully transitioning from conventional to unconventional methods while maintaining safety.
- New logistics and supply chain practices were implemented; water delivery was challenging due to ongoing infrastructure work, but sand delivery processes improved over time.
- Better coordination between drilling and completion teams resulted in smoother well handovers in later stages.
- Effective scheduling was critical to minimizing total completion times.
- Dissolvable fracturing plugs performed well, fully dissolved during mill-out and enhancing downhole operations—30-stage well plug milling could be completed in less than 30 hours
- As activity scaled, demand on surface well testing and flowback equipment increased accordingly.

For upcoming wells, the following recommendations aim to enhance the efficiency and cost-effectiveness of integrated fracturing operations:

- Extend operational parameters, such as increasing lateral length, optimizing fracturing designs, and utilizing more integrated sites.
- Advance technology adoption by leveraging unconventional basin practices or improving local solutions.
- Standardize policies, procedures, contingency plans, and deliverables based on lessons learned to improve project economics without compromising safety.
- Boost operational efficiency in pad-to-pad mobilization and plug-and-perf processes, employing campaign operations when DUC inventory allows.
- Develop a water network for source distribution and wastewater reuse to accommodate growing needs as drilling expands.
- Use locally sourced sand to reduce transportation time and enable further vertical integration for other commodities.
- Ensure adequate production facility capacity to support flowback and hydrocarbon collection.
- Evaluate and implement alternative industry proven toe-initiation methods to streamline preparation.
- Employ surveillance systems during completion for ongoing enhancements, such as casing integrity checks and tracer diagnostics.
- Utilize flexible fracturing and milling equipment setups that can adapt to surface well testing.

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Nomenclature

Bpm	barrel per minute
degF	degree Fahrenheit
Ft	foot/feet
gal	gallon
H ₂ S	Hydrogen Sulfide
ID	inside diameter

lb	pound
OD	outside diameter
PCE	pressure control equipment
ppf	pound per foot
ppg	pound per gallon
psi	pound per square inch
TD	total depth

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